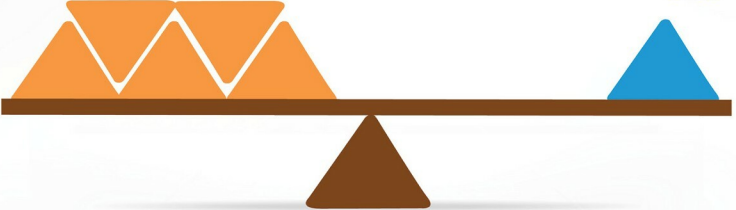


Use Pound on RHEL to Balance the Load on Web Servers



Load balancers distribute network traffic across backend servers and thus enable high availability. Let's see what the process of setting up Pound on Red Hat Enterprise Linux 6.1 x86 is like.

This article explains how to perform software-based load balancing using the Pound package. The Pound server, Web Server1 and Web Server2 are installed on Red Hat Enterprise Linux Server 6.1 (32 bit).

The end user (User1, User2, User3, etc) sends HTTP traffic to the Pound server, which in turn performs the load balancing and sends it to the required backend Web servers.

The default load balancing algorithm used is Round Robin.

Architecture

Shown in Figure 1 is the self-explanatory architectural diagram of the Pound server gateway and the backend Web servers.

Pre-requisites

The following pre-requisites apply to the three servers—the Pound server and both backend Web servers.

1. All servers use RHEL6.1 as OS. The hostnames and IP addresses used are:

```
Pound server    PoundGateway.linuxrocks.org    192.168.10.30
Web server 1    ApacheWebServer1.linuxrocks.org    192.168.10.31
Web server 2    ApacheWebServer2.linuxrocks.org    192.168.10.32
```

2. All servers' /etc/hosts configuration files should contain all the servers' details. They should look like the example shown below:

```
#cat /etc/hosts

127.0.0.1    localhost localhost.localdomain localhost4
localhost4.localdomain4
::1    localhost localhost.localdomain localhost6 localhost6.
```

```
localhost.localdomain
192.168.10.30    PoundGateway    PoundGateway.linuxrocks.org
192.168.10.31    ApacheWebServer1    ApacheWebServer1.linuxrocks.org
192.168.10.32    ApacheWebServer2    ApacheWebServer2.linuxrocks.org
```

3. Set the IP address [Command – ifconfig; System -> Preferences -> Network Connection]
4. Turn off the firewall [System -> Administration -> Firewall]
5. SELinux should be disabled [Command – getenforce; Configuration File – /etc/sysconfig/selinux]

Installation and configuration of Web servers

The default Apache Web Service will be installed on both the backend servers. Since the installation and configuration of Apache Web Service is exactly the same for both the servers, for purposes of documentation, let's record all the steps in ApacheWebServer1. The same steps are to be performed in ApacheWebServer2.

So let's proceed.

Mount the RHEL6.1 DVD and browse the 'Packages' directory.

Install Apache Web Service:

```
[root@apachewebserver1 Packages]# rpm -ivh --aid --force
httpd-2.2.15-9.el6.i686.rpm
warning: httpd-2.2.15-9.el6.i686.rpm: Header V3 RSA/SHA256
Signature, key ID fd431d51: NOKEY
Preparing... #####
##### [100%]
1:httpd #####
##### [100%]
```